

Known-answer validation of an automated Rietveld QPA engine: reproducibility, synthetic recovery, and gate scoring

Paper P-2 of the paper series of the rietveld-agent repository

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August 2026 — working paper, version 0.2.0

Abstract

Thesis. The scientific validity of an automated Rietveld quantitative phase analysis (RQPA) pipeline can be established with three independent checks: (A) bit-level reproducibility across independent runs, (B) known-answer recovery of injected ground-truth fractions on noise-corrupted synthetic patterns, and (C) honest, scored acceptance gates on real data. We report all three for the **rietveld-agent** engine (spike 16 of the repository [3]): (A) two full independent suite runs produced bit-identical scientific payloads (canonical md5 `f43d8be2...`); (B) four synthetic known-answer tests recovered all 24 injected phase rows within ± 1.5 wt% (majors) / ± 1.0 wt% (minors); (C) on the six real refinements the protocol is operational 5/6 and publication-scored 0/6, with the residual wR gap (9.8–27.1% vs. 6.5/5 targets) attributed to microabsorption rather than pipeline failure. The comparison reference for published fractions is a normalized one: the published Table-3 aluminate-residue row sums to 102.6%, and scoring against the raw values was a normalization artifact, now removed. The paper follows the fifteen-step framework of [1].

1 Introduction

A refinement pipeline that runs unattended must earn trust differently from an interactive analyst session. There is no human to notice a plausible-but-wrong fit, and no session log to audit. *Gap.* Published RQPA validation usually stops at “converges and looks reasonable”. What is missing is a three-pronged, executable proof: the results reproduce, the pipeline recovers known truth, and the remaining distance to publication quality is measured and attributed. *Response.* Spike 16 of the **rietveld-agent** repository [3] implements exactly that proof for the protocol of Paper P-1 [3]. Its harness, `benchmarks/spikes/spike_16_validate.py`, runs three independent checks and always completes; verdicts are data, not exit codes.

2 Methods

2.1 A. Reproducibility check

The full spike-15 suite (six refinements, ~ 15 min on the development hardware) is re-run via subprocess. The report’s canonical content hash — computed over all fields except wall-clock timing, which legitimately varies — is compared with the fresh run. Identical hash across independent runs is the reproducibility verdict.

2.2 B. Synthetic known-answer test (KAT)

For each sample (M3 model), the published wt% are *injected* as ground truth, normalized to sum to 100 (the raw Table-3 aluminate-residue row sums to 102.6 owing to replicate-mean rounding;

the normalized reference `published_norm()` is the shared comparison baseline of the repository). Per-phase scales are started from the published-fraction prior via Hill–Howard inversion, $S \sim w/(MV)$ [2], then only the Chebyshev background is refined (three cycles, scales pinned) so the materialized calculated pattern equals the published-fraction phase signal plus a smooth window-adapted background without fit-residual leakage. Poisson counting noise is added at a 200 000-count peak maximum (seed 16) (`benchmarks/eval/noise.py`). The *unmodified* protocol ladder of Paper P-1 then runs on the synthetic pattern from its own starts, with its own constraint/fallback logic. Recovery is scored vs. the injected truth: $|\Delta| \leq 1.5$ wt% for published majors (≥ 5 wt%), ≤ 1.0 wt% for minors.

2.3 C. Gate scoring on real data

Each of the six real refinements is scored against publication-style gates: $wR \leq 6.5\%$ (Cu) / $\leq 5\%$ (synchrotron); $rwp_norm < 1.0$ / 0.5 ; per-phase wt% within the same bands; rank agreement (Kendall τ). *Operational* = converged, no error markers, $wR < 25$. *Publication* = all gates pass. Every instrument, band and gate definition is recorded in the protocol and in `data/spike16/results/spike16_report.md`, together with the probe classification used during development (wheel = procedural choice; counter = counterexample probe; pin = pinned constant; gate = guard on a comparison).

3 Results

3.1 A. Reproducibility

Two independent full-suite runs produced bit-identical payloads: committed run canonical hash and fresh run both `f43d8be2c932420676d612242dd049a5`; cross-run identical = true; rerun rc = 0 (900.5 s); verdict **REPRODUCIBLE**. (One harness bug is worth recording: an earlier version crashed in `write_report` when a killed subprocess was still writing the shared work directory; the writer now renders a failed/skipped rerun explicitly — the verdict is data and the harness always completes.)

3.2 B. Known-answer recovery

All four samples pass; 24/24 phase rows are within band (Table 1); synthetic fits land at the noise floor (wR 0.38–0.56%). The recovery ceiling is ≈ 0.3 wt% — the size of the Poisson noise floor at

sample	KAT score	wR (%)	max $ \Delta $ (wt%)
clinker Cu	8/8	0.45	0.07 (aphthitalite recovered freely)
silicate residue	4/4	0.56	0.10
aluminate residue	5/5	0.38	0.27 (ferrite 67.8 vs. 68.0)
clinker synchrotron	7/7	0.46	0.27 (aluminate 1.80 vs. 1.99)

Table 1: Synthetic known-answer recovery (peak 200 000 counts, seed 16; bands $\pm 1.5/\pm 1.0$ wt%).

this counting statistics. Notable fallouts: aphthitalite recovers freely on the synthetic aluminate window (2.33 vs. 2.44 injected, within band), proving that the degeneracy observed on the real pattern is real-data specific (peak overlap / microabsorption), not intrinsic to the window; and the aluminate ferrite deficit is a normalization artifact of the *reference*, not of the pipeline (67.8 vs. 68.03 normalized injected, $|\Delta| = 0.23$, PASS).

3.3 C. Gate scoring

Operational: 5/6; publication: 0/6, as expected. Every analysis completes, every phase carries a wt%, and the protocol runs deterministically — but the fits are not publication-grade. The

sample	model	wR (%)	wR gate	rank τ	operational
clinker Cu	M3	15.07	FAIL (6.5)	0.36	yes
clinker Cu	T1	20.91	FAIL (6.5)	0.29	yes
silicate residue	M3	20.18	FAIL (6.5)	0.67	yes
silicate residue	T1	27.11	FAIL (6.5)	0.67	no
aluminate residue	M3	13.56	FAIL (6.5)	-0.20	yes
clinker synchrotron	M3	9.76	FAIL (5.0)	0.62	yes

Table 2: Real-data gate scores (rwp_norm gate PASS everywhere; wt gate FAIL; the aluminate residue reports 5/5 phases with aphtthitalite as fixed-composition constraint).

dominant residual is microabsorption: the aluminate-residue fit gained on the real data from the cell + breadth ladder (wR 14.9 $\rightarrow$ 13.56; ferrite 8.4 $\rightarrow$ 17.4 wt%), yet 17.4 vs. 68.0 normalized published remains the largest single-phase gap and is the target of the Brindley-correction spike 17.

4 Discussion and closing

Three results carry the validity argument. *First*, reproducibility is a machine-checked fact, not a claim. *Second*, the engine recovers known fractions to the noise floor of the data on which it was not trained — the KAT design reuses the unmodified production ladder, so the recovery is a property of the pipeline. *Third*, the gates are honest: the publication gap is attributed to physics (microabsorption) with a concrete remedy (spike 17), and the normalization trap (scoring against a reference that sums to 102.6%) was found by the harness and removed by adopting `published_norm()` as the shared comparison baseline. Nothing in this paper claims accuracy the gates do not certify: validity here means “reproduces, recovers known truth, and reports its remaining distance truthfully”. As in Papers P-1 and P-3, the composition follows [1].

License

Apache-2.0; see LICENSE in the repository root. All data and programs needed to reproduce this paper are in the repository [3].

References

- [1] J. M. Drake and B. A. Han. How to write a scientific paper in fifteen steps. *PLoS Computational Biology*, 21(9):e1013505, 2025. doi: 10.1371/journal.pcbi.1013505. PMCID: PMC12459795.
- [2] R. J. Hill and C. J. Howard. Quantitative phase analysis from neutron powder diffraction data using the Rietveld method. *Journal of Applied Crystallography*, 20:467–474, 1987. doi: 10.1107/S0021889887086199.
- [3] qaemu. rietveld-agent: a deterministic rietveld QPA engine for scientific agent runtimes, 2026. URL <https://github.com/qaemu/rietveld-agent-spikes>. Apache-2.0; see docs/papers (series P-1..P-3).